

RUTHVIK VISHWANATHWAR

Senior Data Engineer | Azure Data Engineer | Databricks Engineer

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SUMMARY

Senior Data Engineer with 10+ years of experience modernizing legacy SQL Server/SSIS platforms into Azure Databricks lakehouse architectures. Specialized in PySpark, Delta Lake, ADF, Unity Catalog, and healthcare data engineering, with hands-on experience processing high-volume claims, member, and provider datasets under HIPAA/PHI governance controls. Proven record of reducing pipeline runtimes, improving query performance, and building production-grade data platforms that support analytics, reporting, and operational decision-making.

CERTIFICATIONS

Databricks: Platform Architect, Certified Platform Administrator, Generative AI Fundamentals, AI Agent Fundamentals, Lakehouse Fundamentals | SQL for Data Science

CORE COMPETENCIES

- **Cloud & Platforms:** Azure Data Factory, Databricks, Synapse Analytics, ADLS Gen2, Blob Storage, Logic Apps, DevOps, Microsoft Fabric, OneLake, Snowflake, Power BI
- **Data Engineering:** ETL/ELT Pipelines, Delta Lake, Medallion Architecture, Data Lakehouse, Data Warehousing, SSIS Modernization, Real-Time Streaming (Spark Structured Streaming), Cosmos DB, Lakehouse Federation, Airflow, DAG design, SLA monitoring, backfills, dependency management.
- **Programming:** Python, PySpark, SQL, T-SQL, DAX, Spark SQL
- **Performance & Optimization:** Z-Ordering, Compaction, Caching, OPTIMIZE/MERGE, Data Skipping, Index Tuning, Query Tuning, Cost Optimization
- **AI & ML:** Generative AI, LLM Data Pipelines, MLflow, Feature Engineering, AI Agent Orchestration
- **DevOps & Tooling:** Git, GitHub, Azure DevOps CI/CD, GitHub Actions, Databricks Notebooks, JIRA, Confluence, ServiceNow
- **Governance & Compliance:** HIPAA/PHI Masking, Access Controls, Schema Enforcement, Audit Logging, SCDs, Data Quality Validation

PROFESSIONAL EXPERIENCE

Senior Data Engineer | *Dotcom-Team, Frisco, TX (Clients: Sentara Health, Georgia Pacific)* May 2022 – Present

- Led migration of 40+ legacy SSIS/SQL Server ETL workflows into modular Databricks PySpark/Spark SQL jobs, retiring legacy execution dependencies and reducing average pipeline runtime by ~60% across production workloads.
- Designed reusable Databricks lakehouse engineering patterns for ingestion, transformation, data quality, and deployment, standardizing schema validation, error handling, logging, and replayability across production pipelines.
- Designed and enforced Medallion Architecture (Bronze/Silver/Gold) on Delta Lake, implementing ACID transactions, schema evolution, time travel, and versioning to ensure data reliability and auditability across the entire lakehouse.
- Engineered HIPAA/PHI-compliant data pipelines for large-scale healthcare datasets (claims, member, provider), applying column-level masking, role-based access controls, and PHI tokenization in alignment with enterprise governance standards.
- Drove Delta Lake performance optimization across enterprise datasets using Z-ordering, adaptive partitioning, OPTIMIZE/MERGE, and file compaction, improving query response times by up to 70% on multi-billion row tables.
- Reduced a critical batch process from 14 hours to under 3 hours through index tuning, stored procedure refactoring, and query optimization using SQL Server Database Engine Tuning Advisor.
- Delivered interactive Power BI dashboards and paginated reports for healthcare operations and executive stakeholders, including DAX-driven KPI models and subscription-based delivery.

- Built near-real-time Spark Structured Streaming pipelines for event-driven ingestion into Delta Lake, supporting checkpointing, schema evolution, late-arriving data handling, and replayable processing semantics.
- Standardized Databricks cluster policies and workload-specific compute configurations while optimizing shuffle volume, partition strategy, and redundant transformations to improve cost control, security, and pipeline efficiency.
- Led technical design reviews and mentored data engineers on Spark optimization, Delta Lake modeling, Unity Catalog governance, and CI/CD standards, improving consistency across lakehouse development and production support practices.

Stack: Azure Databricks, PySpark, Spark SQL, ADF, Delta Lake, Delta Live Tables, SQL Server 2019, Azure Synapse, Power BI, SSIS, Python

Senior Data Engineer | *E-IT Professionals, Livonia, MI (Blue Cross Blue Shield of Kansas)* May 2021 – May 2022

- Designed and maintained healthcare reporting data marts using star-schema models, fact/dimension tables, and reusable SQL views to standardize KPI reporting across claims, member eligibility, and provider network analytics.
- Implemented data validation and reconciliation checks across source-to-report pipelines, improving trust in eligibility, claims, and provider metrics by identifying missing records, duplicate claims, and inconsistent member attributes before executive reporting.
- Designed orchestration patterns across ADF, Databricks Jobs, and Airflow DAGs, standardizing retries, dependency handling, SLA monitoring, and backfill operations.
- Optimized Databricks workload efficiency through partition strategy, file compaction, job tuning, and cluster configuration standards, reducing compute waste and improving pipeline scalability across high-volume datasets.
- Tuned high-volume SQL workloads through indexing strategy, execution-plan analysis, and stored procedure refactoring, reducing report refresh time by 40% and improving dashboard responsiveness for finance, claims, and operational reporting users.

Stack: SQL Server 2016/2017, Power BI, Power Query, DAX, GitHub Actions, SSMS, JIRA, Confluence, Databricks, Airflow DAGs

Database Developer | *SkillStorm LLC, Addison, TX (Client: Bank of America)* Nov 2019 – Apr 2021

- Built and deployed SSIS packages for multi-source data extraction and transformation, ingesting from Excel, flat files, and relational databases with Conditional Split, Fuzzy Lookup, and Multicast for data scrubbing and staging validation.
- Designed automated ETL processes ensuring end-to-end data consistency, quality, and scalability across multiple operational systems.
- Investigated and resolved performance bottlenecks in stored procedures and complex SQL queries, improving execution plans and reducing load times across key reporting pipelines.

Stack: SQL Server 2016/2017, SSIS, SSRS, SSAS, Power BI, Visual Studio 2017, JIRA, Confluence, Bitbucket

Database Developer | *E-IT Professionals Corp, Livonia, MI (Client: Visa Inc, Vintech)* Aug 2016 – Oct 2019

- Maintained referential integrity and data consistency using SQL constraints; reviewed and refactored complex multi-join queries to improve performance across high-transaction financial datasets.
- Designed automated ETL processes ensuring end-to-end data consistency, quality, and scalability across multiple operational systems.

Stack: SQL Server 2016, SSIS, SSRS, Visual Studio 2017, JIRA, Confluence, Bitbucket

EDUCATION

M.S. Information Technology | GPA: 3.8/4.00 | Campbellsville University, Kentucky

M.S. Computer Science | GPA: 3.7/4.00 | New York Institute of Technology, Manhattan, NY

B.S. Information Technology | GPA: 3.5/4.00 | Scient Institute of Technology, India